

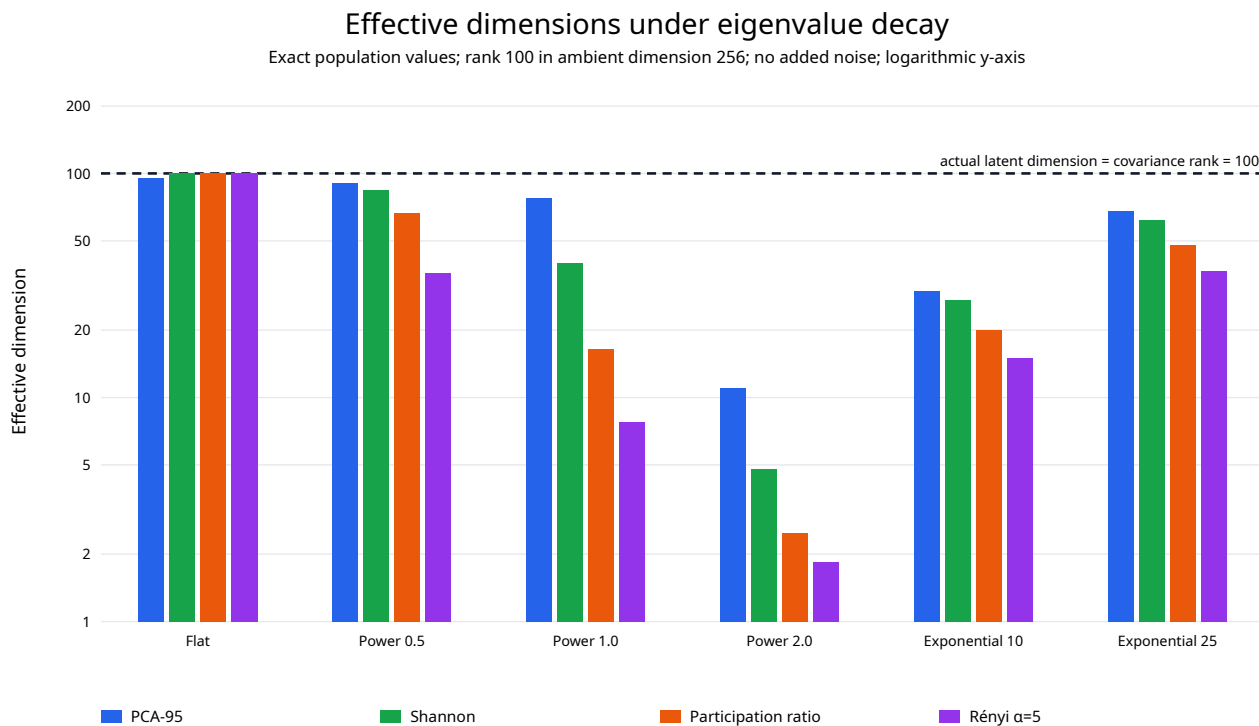
EffDim under eigenvalue decay and ambient noise

Synthetic covariance spectra used rank 100 in 256 ambient dimensions, 10,000 observations, and five trials per condition. Population targets are known exactly and compared with the sample covariance implementation.

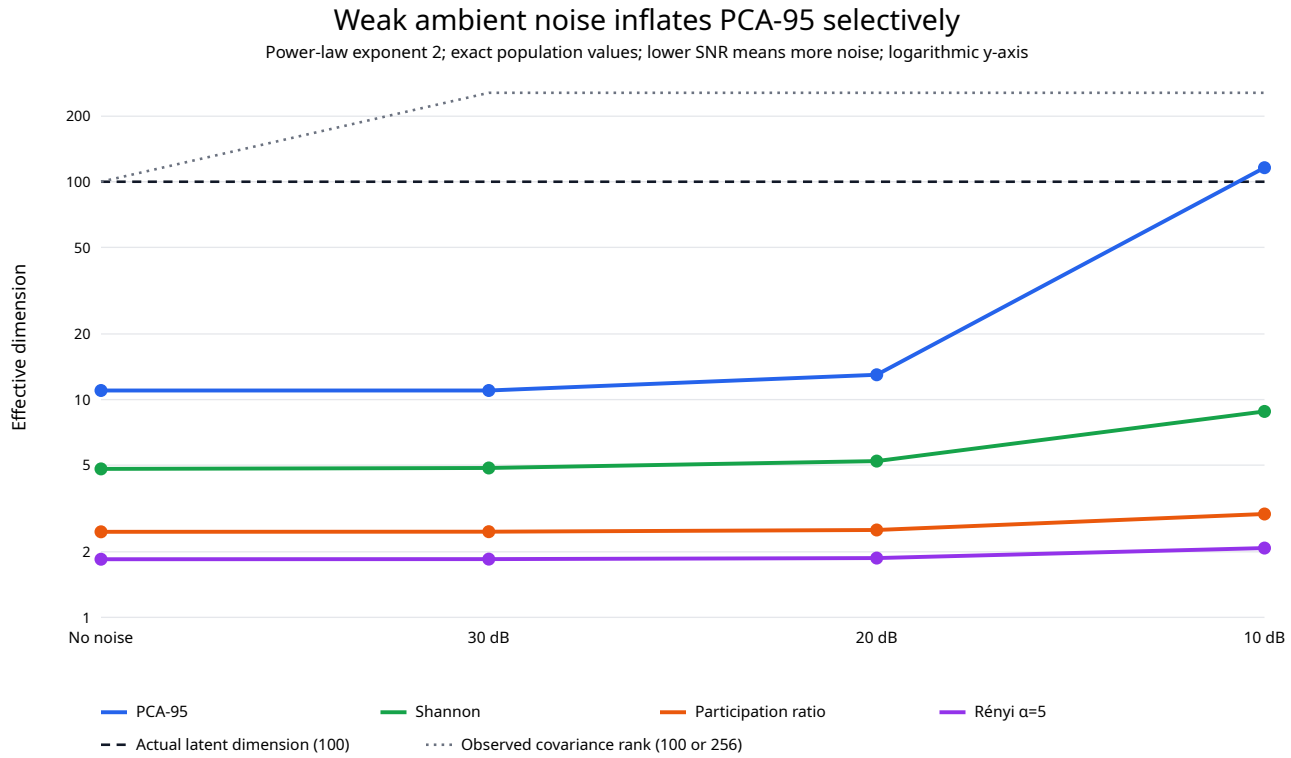
Main findings

1. **The actual latent dimension is 100 in every condition.** Without noise, covariance rank is also 100. Any nonzero isotropic ambient noise makes the population covariance full-rank at 256.
2. **The earlier agreement was a special property of the flat spectrum.** With power-law exponent 1 and no noise, PCA-95 is 78 while Shannon is 39.68, participation ratio is 16.46, and Rényi $\alpha=5$ is 7.76.
3. **Steeper decay increases the disagreement dramatically.** At exponent 2, the same metrics are 11, 4.81, 2.47, and 1.85 even though algebraic rank remains 100.
4. **A weak, broad noise floor can make PCA-95 large without materially changing dominant-direction metrics.** For exponent 2, moving from no noise to 10 dB raises PCA-95 from 11 to 116, but participation ratio only from 2.47 to 2.98 and Rényi $\alpha=5$ from 1.85 to 2.08.
5. **This reproduces the qualitative real-data pattern:** a few dominant directions plus many weak directions can yield high PCA-95 and low Shannon/Rényi dimensions simultaneously.
6. **Sample estimates closely track exact population definitions.** The largest discrepancy is 6.9% for PCA-95 under the noisiest condition; most continuous effective-rank errors are around 0-2.5%.

Spectrum-shape effect



Noise-floor effect



Exact population dimensions without noise

Spectrum	Actual latent d	Covariance rank	PCA-95	Shannon	Participation ratio	Rényi $\alpha=5$
Flat	100	100	95.000	100.000	100.000	100.000
Power 0.5	100	100	91.000	84.412	66.618	35.871
Power 1.0	100	100	78.000	39.677	16.458	7.758
Power 2.0	100	100	11.000	4.808	2.470	1.848
Exponential 10	100	100	30.000	27.181	20.015	14.984
Exponential 25	100	100	68.000	61.919	48.208	36.542

Actual dimension versus effective dimensions under noise

SNR	Actual latent d	Covariance rank	PCA-95	Shannon	Participation ratio	Rényi $\alpha=5$
No noise	100	100	11.000	4.808	2.470	1.848
30 dB	100	256	11.000	4.854	2.475	1.851
20 dB	100	256	13.000	5.216	2.519	1.871
10 dB	100	256	116.000	8.821	2.982	2.081

Sample-to-population accuracy across all 24 conditions

Method	Median relative error	Maximum relative error
PCA-95	0.94%	6.90%
Participation ratio	0.40%	1.16%
Shannon ED	0.44%	0.83%
Rényi $\alpha=2$	0.40%	1.16%
Rényi $\alpha=3$	0.30%	1.56%
Rényi $\alpha=4$	0.26%	2.02%

Method	Median relative error	Maximum relative error
Rényi $\alpha=5$	0.26%	2.47%